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Inter-Individual Variability in Temperature Rise from Localized Exposure at 1–80 GHz Using 131 Anatomical Head Models

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Abstract—International exposure guidelines specify the 10-g averaged specific absorption rate (SAR) below 6 GHz and absorbed power density (APD) above 6 GHz as basic restrictions for local exposure. However, quantitative data on inter-individual variability in radio-frequency (RF)-induced temperature rise remain limited. In this study, temperature rise in the human head was evaluated using 131 anatomical models exposed to near-field RF sources from 1 to 80 GHz. Electromagnetic fields were computed using the finite-difference time-domain method, and steady-state temperature rise was obtained from the bioheat transfer equation. A heating factor (HF), defined as the ratio of the spatial peak temperature rise to 10-g SAR or APD, was used to quantify the relationship between absorbed electromagnetic power and thermal response. Inter-individual variability was substantially lower for APD-based HF than for SAR-based HF. The coefficient of variation remained within approximately 20% for SAR-based HF and below 5% for APD-based HF. Similar behavior was observed across different antenna-to-skin separation distances and skin perfusion conditions. These findings suggest that APD is a stable metric for assessing localized temperature rise above 6 GHz, with only limited influence of anatomical variability.

Index Terms—Radio-frequency exposure, Specific absorption rate (SAR), Absorbed power density (APD), Bioheat modeling, Temperature rise, Anatomical variability

I. INTRODUCTION

THE widespread deployment of wireless communication systems has increased opportunities for human exposure to radio-frequency (RF) electromagnetic fields (EMFs).

For frequencies above 100 kHz, thermal effects are the primary established interaction mechanism between EMFs and biological tissues, and the temperature rise is the basis for local exposure limits specified in the international guidelines established by the International Commission on Non-Ionizing Radiation Protection (ICNIRP) [1] and the Institute of Electrical and Electronics Engineers (IEEE) [2]. To prevent excessive localized tissue heating, these guidelines specify the 10-g averaged specific absorption rate (10-g SAR) below 6 GHz and absorbed power density (APD) above 6 GHz as basic restrictions for localized exposure [1], [2].

To account for uncertainties in the derivation of exposure limits, reduction factors are incorporated into the basic restrictions. For local exposure, these factors are typically 10 for general public and 2 for occupational exposure. They are intended to account for biological and dosimetric variability related to factors such as age, sex, physiological conditions, and environmental conditions, as well as limitations in the available health evidence [1]. Despite their importance, quantitative information on inter-individual variability in RF-induced temperature elevation remains limited, particularly for localized near-field exposure over a broad frequency range spanning both SAR- and APD-based regimes.

Previous computational studies have evaluated SAR or temperature rise using anatomical human models. However, most investigations have been conducted with a simplified model or a limited number of models, often fewer than five [3]–[12]. Representative studies are summarized in Table 1. For example, Morimoto et al. [3] examined the relationship between peak spatially averaged SAR and maximum temperature rise in several head models across GHz frequencies, demonstrating the influence of thermal diffusion and tissue properties at higher frequencies. Other studies employing 15 anatomical models have examined variability in localized or whole-body exposure scenarios [13]. Additional studies have focused on temperature elevation in specific anatomical structures or have conducted cross-species comparisons [14]–[18]. However, the limited number of anatomical models used in previous studies makes it difficult to characterize inter-individual variability.

Since the publication of the revised ICNIRP guidelines and IEEE standard [1], [2], increasing attention has been paid to local exposure above 6 GHz. Many studies have employed simplified models to investigate millimeter-wave exposure [6]–[8], [15], [19]. Although wideband analyses have been reported, such as studies spanning 6–100 GHz using layered tissue models [8], studies based on large populations of realistic anatomical models remain scarce. ICNIRP has noted that a more systematic assessment of age-, sex-, and anatomy-related variability would support future refinement of uncertainty and reduction factors used in exposure assessment [20].

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TABLE I
SUMMARY OF REPRESENTATIVE STUDIES ON LOCALIZED SAR/TEMPERATURE RISE DOSIMETRY USING ANATOMICAL MODELS

Reference	Frequency range	Exposure metric	Model
Morimoto et al. (2016) [3]	1 GHz – 30 GHz	10-g SAR, ΔT	Simplified + Adult 4 + Child 1
Uusitupa et al. (2010) [13]	300 MHz – 5 GHz	10-g SAR, WBA-SAR	Adult 13 (different postures) + Child 2
Hirata et al. (2006) [4]	900 MHz – 2.45 GHz	10-g SAR, 1-g SAR, ΔT	Adult 2
Schmid et al. (2007) [16]	400, 900, 1850 MHz	10-g SAR, ΔT	Local structures (eye, inner ear, pineal gland)
Kodera and Hirata (2018) [14]	3–10 GHz	10-g SAR, ΔT	Adult 1 (vs. rat model)
Morelli et al. (2021) [5]	28 GHz	10-g SAR, APD	Adult 2 + Child 2
Sacco et al. (2021) [6]	26, 60 GHz	APD, ΔT	Simplified age-dependent tissue model (ages 5–70)
Nakae et al. (2020) [7]	28 GHz	IPD, APD, ΔT	Simplified model
Haider et al. (2023) [15]	60 GHz	APD, ΔT	Microscale nerve/capillary model
Christ et al. (2020) [8]	6–100 GHz	IPD, ΔT	Simplified model
This study	1–80 GHz	10-g SAR / APD (4, 1 cm ²), ΔT	Adult 131 (50 male, 81 female)

In this study, we evaluated localized RF-induced temperature rise in the human head using 131 anatomical models derived from magnetic resonance (MR) images. The analysis spans frequencies from 1 to 80 GHz, covering the frequency ranges where SAR and APD are applied as exposure metrics. To quantify inter-individual variability, a heating factor (HF) is defined as the ratio of the spatial peak temperature rise to the corresponding exposure metric. The distributions of this quantity are examined as functions of frequency, age, and sex, providing a population-based assessment of variability in localized RF-induced heating across a wide frequency range.

II. MODELS AND COMPUTATIONAL METHODS

A. Anatomical Human Head Models

We employed 131 anatomical head models constructed from MR images [21]. These models were obtained from the open-access dataset named Segmented Head Anatomical Reference Models (SHARM) [21]. The SHARM was developed from open-access multi-sequence MR images (available at <https://brain-development.org/ixi-dataset/>) with a variety of genders, ages, body mass indices (BMI), and racial backgrounds. The models comprise 50 adult males and 81 adult females (mean age: 44.7±16.1 years). Figure 1 presents representative voxel-based head models used in this study.

The models were generated from T1- and T2-weighted MR images using a semi-automatic segmentation procedure [22]. As shown in Fig. 2, the processing pipeline consisted of image preprocessing, tissue segmentation using ForkNet [23], and anatomical model construction within the SHARM framework for electromagnetic computation [21]. Each model included 15 tissue types: skin, muscle, fat, cortical bone, trabecular bone, cerebrospinal fluid, blood vessels, dura mater, cerebral gray matter, cerebral white matter, cerebellar gray matter, cerebellar white matter, vitreous body, lens, and mucosal tissue.

The original voxel resolution was 1 mm. To satisfy numerical stability requirements [24] and reduce discretization error [25],

surface smoothing was applied. The spatial resolution was refined to 0.5 mm for 1–10 GHz, 0.25 mm for 15–30 GHz, and 0.125 mm for 40–80 GHz to ensure that the discretization size was less than one-tenth of the wavelength in tissue at each frequency [26], [27]. Due to the limited contrast in MR images, the boundary between the skin and subcutaneous fat was sometimes indistinct. To improve consistency among models, we reassigned the outermost layer voxels of each model using average tissue thicknesses based on [28]. Specifically, we manually segmented the superficial layers from the outside in as skin (1.5 mm), fat (1.5 mm), and muscle (2.5 mm).

B. Electromagnetic Field Computation

Electromagnetic fields were computed using the finite-difference time-domain (FDTD) method [16]. A 10-layer convolutional perfectly matched layer (CPML) was implemented as the absorbing boundary condition to suppress boundary reflections from the computational domain boundaries [17]. Frequency-dependent dielectric properties of biological tissues were modeled using the four-pole Cole–Cole dispersion model [18].

The SAR was calculated from the local electric field as

$$SAR(\mathbf{r}) = \frac{\sigma(\mathbf{r})}{2\rho(\mathbf{r})} |\mathbf{E}(\mathbf{r})|^2, \quad (1)$$

where $\mathbf{E}(\mathbf{r})$ is the maximum electric field magnitude at position $\mathbf{r}$, $\sigma(\mathbf{r})$ is the electrical conductivity, and $\rho(\mathbf{r})$ is the mass density of the tissue.

The 10-g averaged SAR (SAR_{10g}) was obtained by spatial averaging over a cubic volume corresponding to 10 g of tissue in accordance with IEEE recommendations [29].

For frequencies above 6 GHz, APD was computed by integrating the absorbed power over the tissue volume underlying a specified surface area and dividing it by that area. Averaging areas of 4 cm² and 1 cm² were considered (APD_{4cm^2} , APD_{1cm^2}), consistent with current exposure guidelines [1], [2], with the 1 cm² area applied above 30 GHz.

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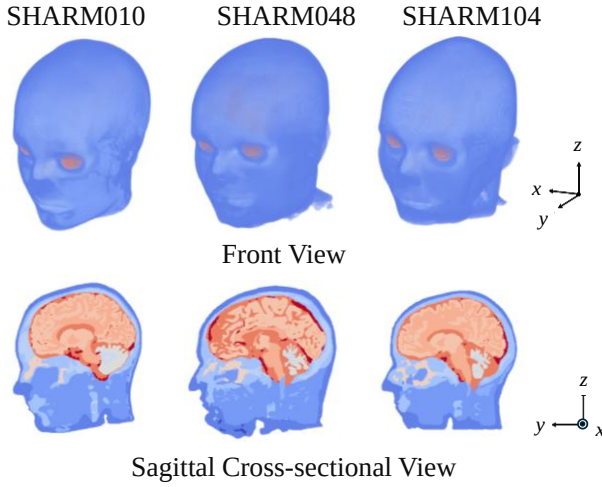


Fig. 1. External view and sagittal cross-section of the representative anatomical head models used in this study.

C. Temperature Rise Computation

Temperature rise in tissue was computed by solving the Pennes bioheat transfer equation [30]. Under continuous-wave exposure, the equation can be reformulated in terms of steady-state temperature rise as follows [26]:

$$\nabla \cdot (K(\mathbf{r})\nabla\Delta T(\mathbf{r})) + \rho(\mathbf{r})SAR(\mathbf{r}) - B(\mathbf{r})\Delta T(\mathbf{r}) = 0, \quad (2)$$

where $\Delta T(\mathbf{r})$ is the temperature rise, $K(\mathbf{r})$ is the thermal conductivity, and $B(\mathbf{r})$ represents the perfusion-related heat loss term.

At the skin-air interface, heat exchange with the environment was modeled using a convective boundary condition as follows:

$$K(\mathbf{r})\frac{\partial}{\partial \mathbf{n}}\Delta T(\mathbf{r}) + H\Delta T(\mathbf{r}) = 0, \quad (3)$$

where $\mathbf{n}$ is the normal component at the skin surface, and H is the convective heat transfer coefficient. The ambient temperature was assumed to be spatially uniform.

Thermal properties were adopted from established literature sources [31], [32], and the blood perfusion coefficient for skin tissue was taken from [33] ($B = 7367 \text{ W/m}^3/\text{°C}$). In the sensitivity analysis of perfusion effects, the skin blood perfusion rate was varied with reference to values reported in [34] ($B = 3687 \text{ W/m}^3/\text{°C}$). Transient thermoregulatory responses were not considered because the localized temperature elevations examined in this study are not expected to significantly affect systemic physiology [35].

The resulting system was discretized using a finite-difference scheme and solved with a geometric multigrid method to handle the large number of anatomical models [26]. The numerical framework presented in Sec. II-B and II-C has been validated against experimental data and intercomparison studies in [36]–[41].

D. Exposure Configuration

A half-wave dipole antenna operating over the 1–80 GHz frequency range was used as a localized exposure source. The antenna was placed adjacent to the lateral side of the head model near the temporal region, with its axis perpendicular to the skin surface, as shown in Figure 3. The half-wave dipole antenna was modeled using a delta-gap source fed across a single-cell feed gap (gap length = 1 cell, corresponding to the spatial resolution at each frequency). The parameters were adjusted to ensure a reflection coefficient (S_{11}) of less than -10 dB . Similar half-wave dipole configurations have been used in previous studies of near-field exposure [42]. The antenna-to-

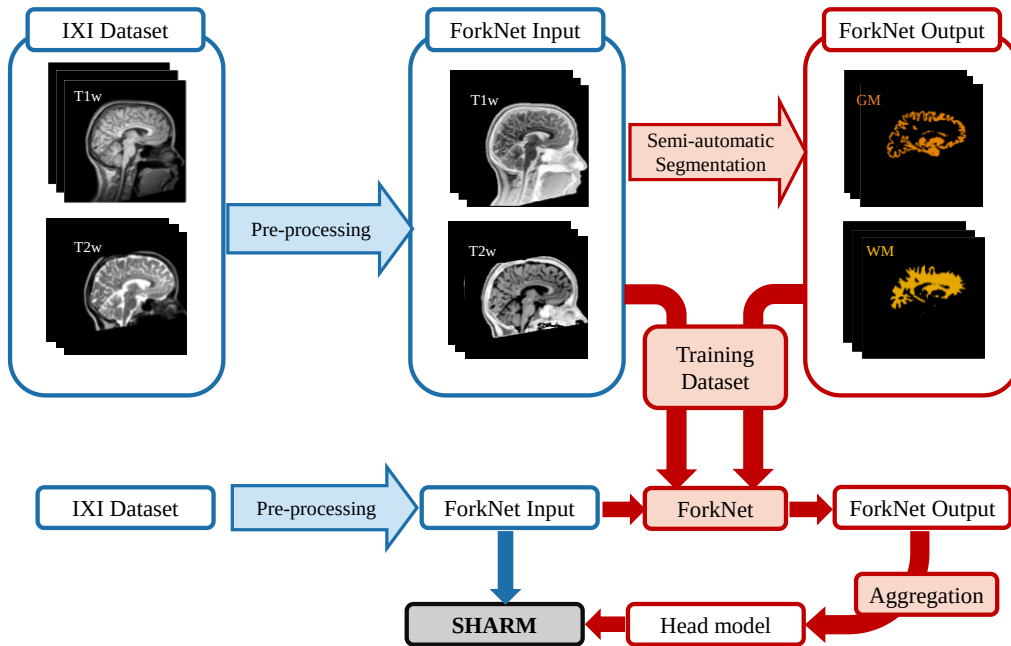


Fig. 2. Processing pipeline for constructing statistical head models from MR images, including preprocessing, tissue segmentation using ForkNet, and model generation within the SHARM framework.

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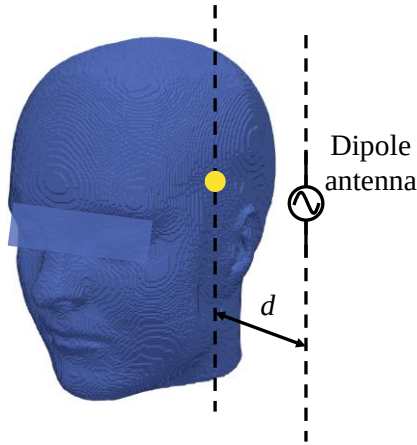


Fig. 3. Exposure configuration. A half-wave dipole antenna is positioned at a distance d from the lateral surface of the head model with its axis perpendicular to the skin surface.

skin separation distance d varied from 5 mm to 25 mm in 5 mm increments. All simulations were performed under continuous-wave exposure with an input power of 20 dBm (100 mW).

E. Heating Factor Definition

To characterize the relationship between electromagnetic power absorption and the resulting thermal response, HF was introduced. Similar HFs have been used in previous dosimetric studies to relate absorbed electromagnetic power to temperature rise [37], [43], [44].

For frequencies below 6 GHz, HF was defined as

$$HF_{SAR} = \frac{\Delta T_{peak}}{SAR_{10g}}, \quad (4)$$

where ΔT_{peak} denotes the spatial peak temperature rise within the tissue.

For frequencies above 6 GHz, HF was defined as

$$HF_{APD} = \frac{\Delta T_{peak}}{APD_{4cm^2, 1cm^2}}. \quad (5)$$

Boxplots show the median (center line), interquartile range (IQR), and whiskers extending to the minimum and maximum non-outlier values. Outliers were defined as observations more than $1.5 \times IQR$ beyond the quartiles. Inter-individual variability was quantified using the coefficient of variation (CV) of HF across the 131 models.

III. RESULTS

A. Frequency Characteristics of Power Absorption and Temperature Rise

Figure 4 presents representative spatial distributions of SAR and temperature rise at 1, 6, and 10 GHz for an antenna-to-skin separation distance of 15 mm. As frequency increases, the SAR becomes increasingly concentrated in superficial tissues owing to the reduced penetration depth. The corresponding temperature rise distribution extends over a broader region due to thermal diffusion, resulting in smoother spatial gradients than those of SAR. The differences in the distribution of the absorption region were observed among the three models, which can be attributed to variations in head geometry. While these geometric differences lead to variations in the spatial pattern of SAR, the corresponding variations are attenuated in the temperature rise distribution.

Figure 5 shows the frequency dependence of SAR_{10g} , APD_{4cm^2} , APD_{1cm^2} , and the spatial peak temperature rise at the same antenna distance. Because the exposure configuration was identical for all models, the observed variability reflects anatomical differences. The SAR_{10g} , evaluated between 1 and 6 GHz, reaches its maximum around 2 GHz and remains within approximately 1.0–1.4 W/kg in the 3–6 GHz range. Within its applicable frequency range, SAR_{10g} shows greater variability

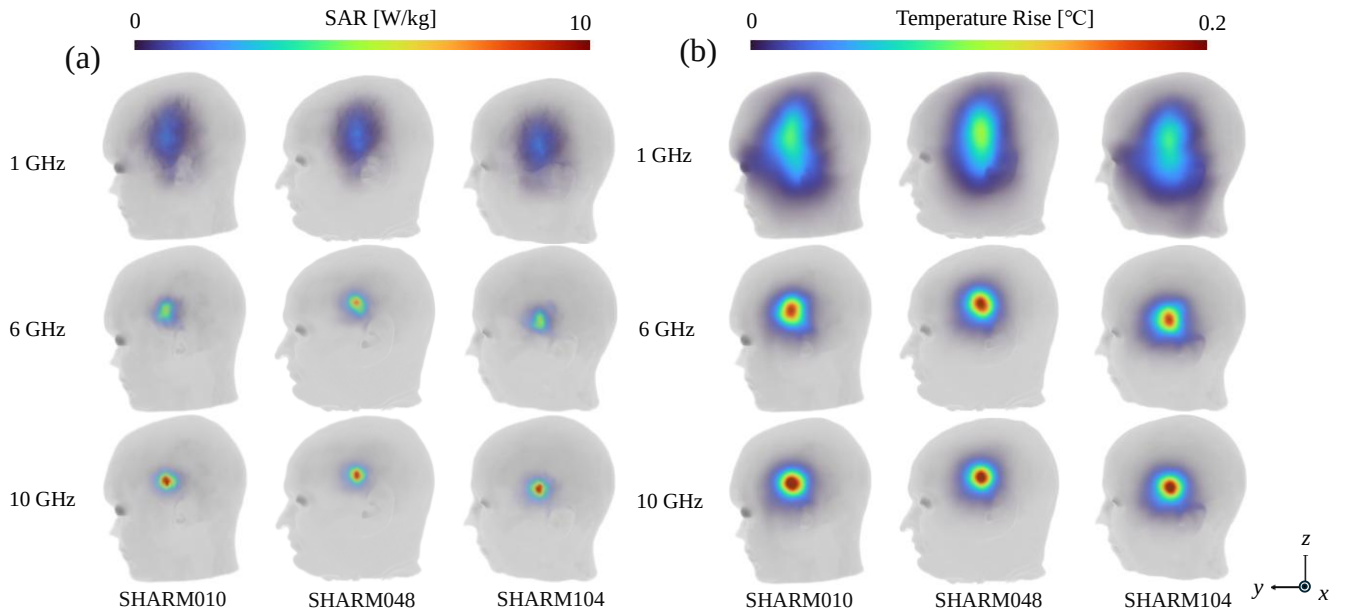


Fig. 4. Representative spatial distributions of local SAR and temperature rise at 1, 6, and 10 GHz at an antenna-to-skin separation distance of 15 mm (input power: 20 dBm).

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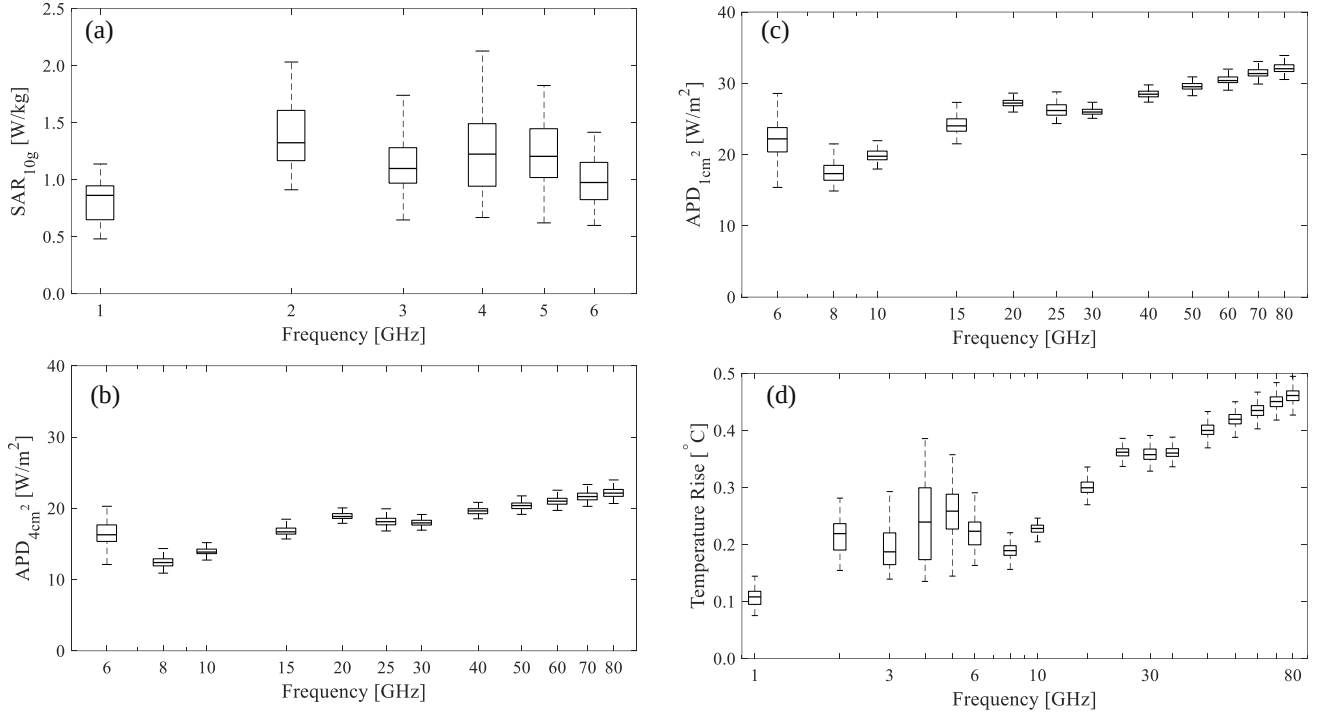


Fig. 5. Frequency dependence of (a) SAR_{10g}, (b) APD_{4cm²}, (c) APD_{1cm²}, and (d) spatial peak temperature rise at an antenna-to-skin separation distance of 15 mm (input power: 20 dBm). Box plots represent variability across 131 anatomical models. The box indicates the interquartile range, the center line indicates the median, and the whiskers extend to the minimum and maximum non-outlier values.

across the models. In contrast, APD_{4cm²} and APD_{1cm²} increase with frequency across 6–80 GHz. Although both averaging areas are shown for comparison, the current exposure guidelines apply the 1 cm² averaging area only above 30 GHz [1], [2]. Spatial peak temperature rise increases gradually with frequency. Larger differences among anatomical models are observed at lower frequencies, whereas the spread across models becomes smaller in the millimeter-wave range.

B. Inter-Individual Variability of Heating Factor

Figure 6 shows the frequency dependence of HF. As shown in Fig. 6, the SAR_{10g}-based HF increases from 1 to 6 GHz. In contrast, the APD-based HF increases from 6 GHz to around 30 GHz and remains nearly constant between 30 and 80 GHz. Similar trends are observed for both the 4 cm² and 1 cm² averaging areas. At higher frequencies, HF based on the 1 cm² averaging area shows slightly less frequency dependence.

Figure 7 shows the CV of HF across the 131 anatomical head models. For SAR_{10g}-based HF, the CV remains relatively high throughout the frequency range, reaching a maximum of 19% at 1 GHz. The CV decreases from 1 to 4 GHz and increases again above 4 GHz. The CV of APD-based HF is lower than that of SAR_{10g}-based HF across most frequencies. In particular, it remains nearly constant at approximately 5% above 20 GHz. Comparable CV values are obtained for APD_{4cm²}- and APD_{1cm²}-based HF, although slightly lower values are observed for APD_{1cm²}.

C. Sensitivity to Exposure Geometry and Blood Perfusion

Figure 8 shows the HF as a function of antenna-to-skin

separation distance d for representative frequencies. As shown in Fig.8(a), the SAR_{10g}-based HF increases as the antenna approaches the skin surface, and the variability across models becomes larger at shorter distances. When the separation distance exceeds 20 mm, the HF shows only small changes across frequencies. As shown in Fig. 8(b), the APD_{4cm²}-based HF is substantially higher at $d = 5$ mm compared to larger separation distances, and the variability among models is also greater. In contrast, the APD_{1cm²}-based HF remains nearly constant across all antenna-to-skin separation distances examined, as shown in Fig. 8(c).

Figure 9 shows HF for different skin blood perfusion rates. Under reduced-perfusion conditions, HF increases across the entire frequency range compared with the baseline condition, with a maximum increase of approximately 9% for SAR_{10g}-based HF. APD-based HF also increases when perfusion is reduced. The maximum increase is approximately 7%, which is smaller than that observed for SAR_{10g}-based HF.

IV. DISCUSSION

This study evaluated temperature rise in the human head for local exposure using 131 anatomical models. The analysis covered frequencies from 1 to 80 GHz and enabled assessment of inter-individual variability across both SAR- and APD-based exposure regimes.

As shown in Fig. 6, the frequency dependence of HF differed between SAR- and APD-based HF and was consistent with previous findings [45], [46]. The SAR_{10g}-based HF increases

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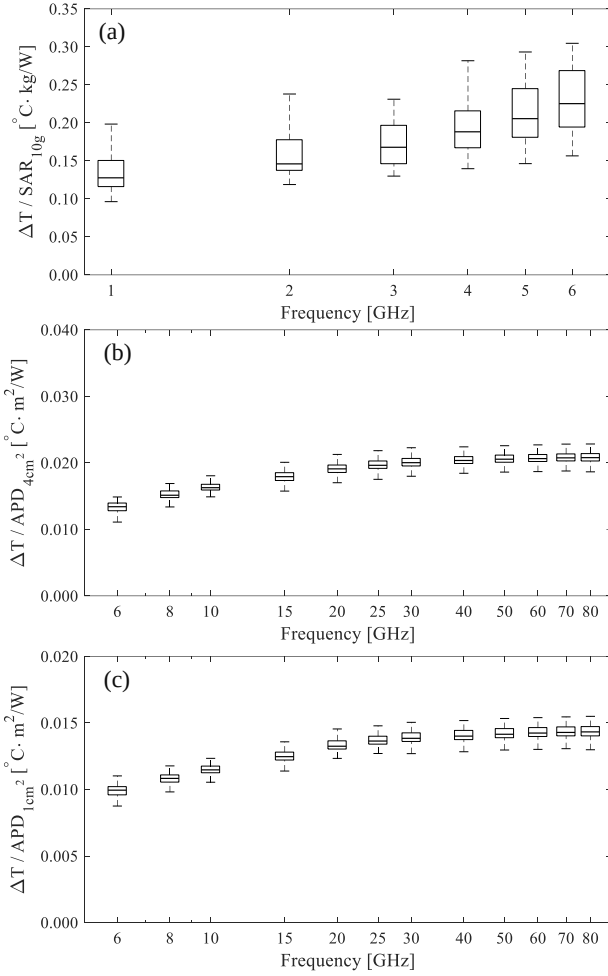


Fig. 6. Frequency dependence of heating factor based on (a) SAR_{10g}, (b) APD_{4cm²}, and (c) APD_{1cm²} at an antenna-to-skin separation distance of 15 mm. Box plots represent variability across 131 anatomical models. The box indicates the interquartile range, the center line indicates the median, and the whiskers extend to the minimum and maximum non-outlier values.

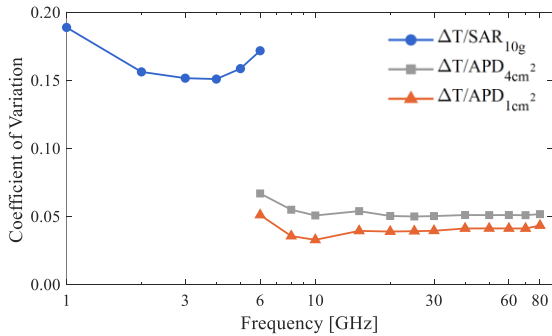


Fig. 7. Coefficient of variation of heating factor based on SAR_{10g}, APD_{4cm²}, and APD_{1cm²} in Fig. 6.

with frequency within its applicable range. This increase is associated with the reduced penetration depth at higher frequencies. The 10-g averaging mass corresponds approximately to a cube with side lengths of about 2 cm, which

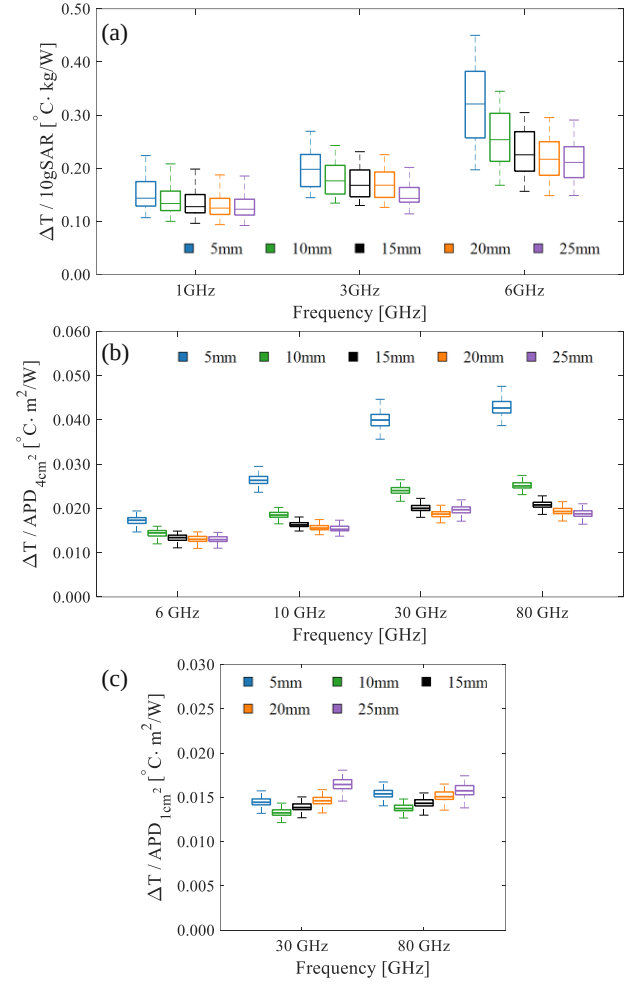


Fig. 8. Dependence of heating factor on antenna-to-skin separation distance d for (a) SAR_{10g}, (b) APD_{4cm²}, and (c) APD_{1cm²}. Box plots represent variability across 131 anatomical models. The box indicates the interquartile range, the center line indicates the median, and the whiskers extend to the minimum and maximum non-outlier values.

is comparable to the thermal diffusion length [3], [47]. At higher frequencies, electromagnetic energy is deposited predominantly in superficial tissues. As power absorption becomes increasingly concentrated near the surface, the relationship between 10-g volumetric averaging and spatial peak temperature rise becomes less direct [45]. In contrast, APD-based HF approaches a nearly constant value above approximately 30 GHz, indicating a more stable relationship between absorbed power and temperature rise at millimeter-wave frequencies.

The observed variability of HF can be compared with the uncertainties addressed by reduction factors adopted in the international guidelines [1], [2]. As shown in Fig. 7, the CV remains within approximately 20% for SAR_{10g}-based HF and below 5% for APD-based HF. These values are substantially smaller than the reduction factor of 10 applied for general public and 2 for occupational exposure. This suggests that anatomical variability represents only one component of the

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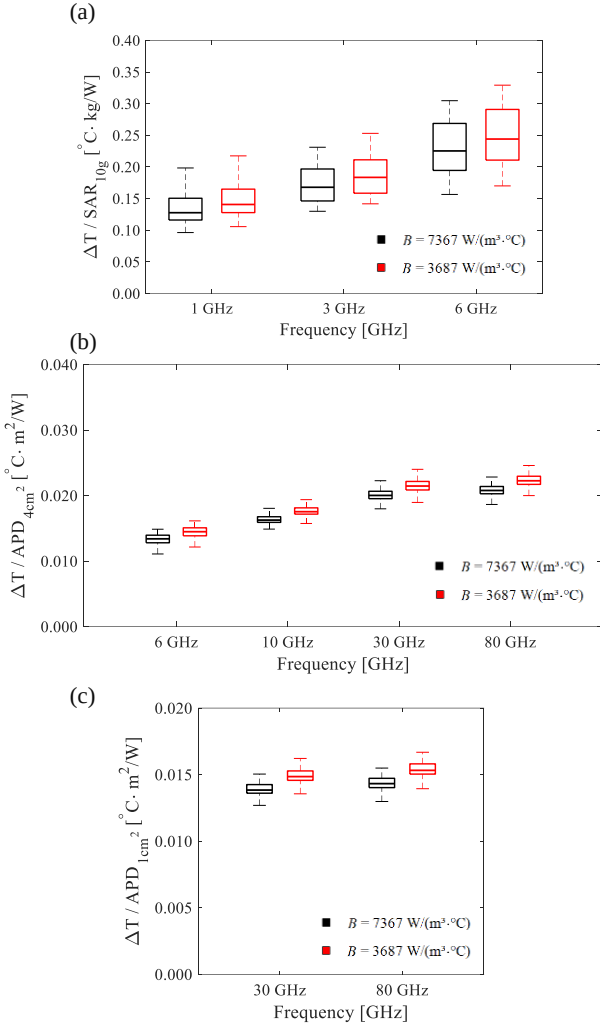


Fig. 9. Dependence of heating factor on skin blood perfusion for (a) SAR_{10g}, (b) APD_{4cm²}, and (c) APD_{1cm²}. Box plots represent variability across 131 anatomical models. The box indicates the interquartile range, the center line indicates the median, and the whiskers extend to the minimum and maximum non-outlier values.

uncertainty considered in guideline development.

Population-based analysis is not the only approach for characterizing dosimetric variability. Statistical simulation methods, such as the unscented transform, have been used to estimate SAR variability from a limited number of deterministic sample points representing perturbations in tissue properties and body geometry [48]. Such approaches are efficient for evaluating the influence of individual parameters, whereas the present analysis captures variability arising from realistic anatomical differences across a large population of anatomical models. An important next step is to combine the population-based dataset employed in this study with non-intrusive stochastic techniques so that anatomical variability and uncertainties in dielectric, thermal, and physiological tissue parameters can be treated simultaneously. This would allow the relative contributions of these factors to the variability of the heating factor to be quantified and may provide further insight

into the rationale behind the reduction factors adopted in the international guidelines [1], [2].

The sensitivity analysis of antenna-to-model distance (Fig. 8) shows that SAR_{10g}-based HF varies substantially at short distances ($d = 5$ mm), where both the HF and inter-individual variability increase. This behavior can be attributed to near-field interactions at short distances, resulting in greater sensitivity to anatomical differences among models.

For APD_{4cm²}, HF also increases substantially at short distances ($d = 5$ mm), particularly at higher frequencies, as shown in Fig. 8(b). This increase can be attributed to the combined effects of source proximity and increasingly localized power absorption at higher frequencies. Under these conditions, averaging over a 4 cm² area may not fully capture the concentrated absorption, resulting in an increase in HF. In contrast, when the antenna-to-model distance exceeded approximately 15 mm, inter-individual variability in HF decreased. This suggests that the anatomical differences and source proximity decrease with increasing separation distance.

HF based on APD_{1cm²} remains nearly independent of antenna-to-model distance (Fig. 8(c)), with little variation across the distances investigated. This result suggests that the 1 cm² averaging area reduces the influence of exposure geometry on HF at short antenna-to-model distances. Because the 1 cm² averaging area is applied above 30 GHz in current exposure guidelines, the limited distance dependence observed for APD_{1cm²}-based HF supports its use in the millimeter-wave region.

As shown in Fig. 9, a reduction in skin blood perfusion increased HF across all exposure metrics, consistent with reduced heat removal in superficial tissues. However, even when blood perfusion was halved, the increase remain within approximately 9% for SAR_{10g}-based HF and approximately 7% for APD-based HF. At millimeter-wave frequencies, perfusion effects did not substantially alter the frequency dependence of APD-based HF. The limited changes in HF suggest that, under the local exposure conditions considered here, geometric factors play a more important role than small variations in skin perfusion.

Although all simulations in this study were performed using a single validated FDTD solver, differences among numerical solvers and implementations may introduce additional variability. The intercomparison studies using a planar model reported relative standard deviations (RSDs) of approximately 8–19% for a single-dipole exposure scenario over 10–90 GHz [39]. This suggests that variability associated with numerical implementation can be comparable to, or slightly larger than, the anatomical variability quantified in this study.

The antenna type is another important source of uncertainty for near-field exposure assessments. The exposure metrics derived from antenna configurations with multiple antenna elements, which are commonly used in practical 5G/millimeter-wave devices, tend to show greater variability than those obtained with a single dipole source [39], [41]. This tendency was particularly pronounced under extreme conditions around 10 GHz, with the RSD generally decreasing as frequency

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increased.

These observations are consistent with the findings of this study. As shown in Fig. 8, the inter-individual variability becomes larger under extreme proximity conditions, suggesting that localized field distributions strongly influence the resulting thermal response. Because realistic patch and array antennas produce more complex and spatially non-uniform near-field distributions than a single dipole, they may result in greater variability.

For example, a principal component analysis performed using realistic human face models showed that antenna configuration (single dipole vs. 4×4 array) was the dominant contributor to variability, accounting for approximately 94% of the total variance, whereas the contribution of anatomical skin-model differences was relatively small [40]. Therefore, while the overall frequency-dependent trends observed in this study are expected to remain qualitatively valid, the CV may become larger when multiple antenna elements are considered.

Taken together, these findings suggest that all sources of variability increase under extreme proximity conditions. Although variability due to differences in numerical methods may be comparable to inter-individual anatomical variability, antenna configuration may represent the dominant source of variability in specific scenarios. These factors should also be considered when evaluating the overall uncertainty in population-based thermal dosimetry.

One limitation of this study is the simplified representation of superficial tissues. To improve consistency among models, the outermost tissue layers were reassigned using fixed thicknesses for skin, fat, and muscle based on reference anatomical data. Although this procedure reduced uncertainty associated with segmentation of superficial tissues in MR images, it may have reduced part of the inter-individual variability in superficial anatomy. Further studies incorporating subject-specific superficial tissue structures may help clarify their influence on HF.

V. CONCLUSION

This study evaluated temperature rise in the human head using 131 anatomical models over the frequency range from 1 to 80 GHz. APD-based HF demonstrated lower inter-individual variability and reduced sensitivity to exposure geometry than SAR_{10g}-based HF.

The results showed that anatomical differences among adult head models produced bounded variations in temperature rise. In particular, APD-based HF showed less than 5% variability above 6 GHz, indicating stable scaling between absorbed power and temperature rise in the millimeter-wave region. These findings support the use of APD for assessing thermal effects under local exposure conditions above 6 GHz.

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